

This completes Squid's configuration. Under *Real Time* you can view the current Squid Access, Squid Cache tables and if enabled, SquidGuard and ClamAV tables too.

Installation and configuration of SquidGuard

Go back to the package manager and install SquidGuard. Successful installation will add the *SquidGuard Proxy Filter* menu item under *Services*. Figure 6 details the required configuration items.

For SquidGuard to filter out the unnecessary websites, a blacklist will have to be configured and updated periodically. This is possible from *Services – SquidGuard Proxy Filter – Blacklist options*. One of the most widely used blacklist URL is <http://www.shallalist.de/Downloads/shallalist.tar.gz>. Please read the terms and conditions of using the Shallalist website beforehand.

Before continuing further, please go through the steps detailed at the end of this article to overcome the known issues associated with Squid and SquidGuard.

Continue to update the blacklist from the *Services – SquidGuard Proxy Filter – Blacklist* menu by clicking on the *Download* button. Once this download is completed, as a primary configuration, go to the *Services – SquidGuard Proxy Filter – Common ACL* menu, select the *Expand Target Rules Lists* by clicking the '+' button, deny access for unwanted target categories such as gambling, porn, etc. Then click on *Save*. See Figure 7 for this configuration example.

Once configured, implement this configuration by clicking the 'Apply' button from the *Services – SquidGuard Proxy Filter – General* menu.

Verify that the filtering is working properly by opening one of the blocked sites in the Web browser. The browser should give a *Request denied* error along with the reason and client IP address. This unauthorised access attempt is also logged and can be viewed from *SquidGuard Proxy Filter – Logs*.

Table 1: Important pfSense menu items

Enable SSH access	<i>System – Advanced – Admin Access – Secure Shell – Enable Secure Shell</i> . If required, you can define the non-standard SSH port number.
Shutdown/restart	<i>Diagnostic – Halt/Diagnostic – Restart</i>
Check running services	<i>Status – Services</i>
Backup & Restore settings	<i>Diagnostics – Backup & Restore</i>
Reinstall existing packages	
Checking Interface Status	<i>Status – Interfaces</i>
Connect/Disconnect WAN	

Overcoming known Squid proxy problems

Detailed instructions and discussions related to the

Request denied by pfSense proxy: 403 Forbidden

Reason:

Client address: 192.168.1.100
 Client name: 192.168.1.100
 Client group: default
 Target group: blk_BL_porn
 URL: http://www.scx.com/

Figure 8: SquidGuard request denied

issue of SquidGuard not able to auto restart after system reboot are available at: <https://forum.pfsense.org/index.php?topic=94312.0>

For the ready reference of users, the steps from this discussion are detailed below.

1. Login to your pfSense computer using SSH and monitor cache log:

```
tail -f /var/squid/logs/cache.log
```

2. In the proxy filter SquidGuard/target categories, define a dummy custom target category. *Name*: Dummy. *Description*: Dummy custom target category (Fix: Squid and SquidGuard are not set to auto start after reboot). The remaining parameters are blank.
3. In the proxy filter SquidGuard/Common Access Control List, in *Target Rules*, find your dummy target category and set access to *Deny* to create the missing blacklist directory.
4. Go to *General Settings* and select *Apply* to activate the configuration update.
5. Your monitoring tail of *cache.log* will immediately start to scroll, indicating successful implementation of the workaround.
6. It is advisable to do a configuration backup (diagnostics/backup/restore).
7. Do a system restart and validate that Squid and SquidGuard services now successfully auto-start. 

References

- [1] Download: <https://pfsense.org/download/mirror.php?section=downloads>
- [2] Hardware requirements: <https://www.pfsense.org/hardware/#requirements>
- [3] Squid not restarting: <https://forum.pfsense.org/index.php?topic=94312.0>
- [4] SquidGuard configuration: https://doc.pfsense.org/index.php/SquidGuard_package
- [5] Shalla blacklist: <http://www.shallalist.de/Downloads/shallalist.tar.gz>
- [6] New features and changes: https://doc.pfsense.org/index.php?title=2.3_New_Features_and_Changes

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